Sri Lanka Institute of Information Technology



IT3091 – MACHINE LEARNING

Decision Log

Group Number: 2026-DS-35

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Decision Area	Decision	Reason
Business objective	Predict late-delivery risk before dispatch	Supports proactive operational intervention.
Unit of analysis	One order/shipment record	Matches the notebook's decision framing.
Target	Late_delivery_risk	Already supplied as a binary 0/1 label.
Secondary lens	Delay pattern analysis	Used to explain patterns by shipping mode, market and category; not a second prediction target.
Class imbalance	No SMOTE/class rebalancing	Target is relatively balanced: 54.83% late vs 45.17% not late.
Leakage control	Drop Delivery Status	It perfectly matches/reveals the target.
Leakage control	Drop Days for shipping (real)	Actual shipping time is only known after shipment and is used to determine lateness.
Leakage control	Drop shipping date (DateOrders)	Post-order information closely tied to delivery outcome.
Privacy	Drop names, email, password, street and ZIP fields	Avoid PII/sensitive fields and non-generalizable information.
Identifiers	Drop order/customer/product/category/department ID fields and Product Status	Identifiers do not represent stable predictive patterns.
Missing data	Drop Product Description and Order Zipcode	100% missing and ~86% missing respectively.
Date handling	Extract order_month and order_dayofweek	Preserves possible seasonal/weekly patterns without using raw timestamp.
Split strategy	80/20 stratified train-test split; random_state=42	Preserves class proportions and reproducibility.
High-cardinality encoding	Smoothed target encoding, k=20	Avoids thousands of one-hot columns; rare categories shrink toward overall rate.
Leakage-safe encoding	Learn target-encoding maps from training set only	Prevents test information from leaking into training.
Unknown categories	Fallback to overall training late-delivery rate	Allows unseen test categories to be scored safely.
Low-cardinality encoding	One-hot encoding with drop_first=True	Appropriate for nominal categories with manageable cardinality.
Scaling	StandardScaler only for Logistic Regression	Linear coefficient model benefits from scaling; tree models do not require it.
Baseline model	Logistic Regression	Simple, fast and interpretable reference point.
Alternative models	Decision Tree, Random Forest, XGBoost	Tests non-linear rules, ensembles and boosting on tabular data.
Decision Tree controls	max_depth=16, min_samples_leaf=8	Limits memorization/overfitting while retaining interpretable rules.
Random Forest controls	300 trees, max_depth=18, min_samples_leaf=3	Balances predictive power and overfitting control.
XGBoost controls	350 trees, depth=7, learning_rate=0.1,	Strong tabular model with sampling

	subsample=0.9, colsample_bytree=0.9	regularization.
Evaluation metrics	Accuracy, Precision, Recall, F1, ROC AUC	Gives both threshold-based and ranking-quality views.
Robustness check	3-fold stratified CV on 40,000-row training sample	Checks stability while keeping runtime reasonable.
Operational recommendation	XGBoost primary; Random Forest backup; Decision Tree for explanation	XGBoost has best test Accuracy, F1 and ROC AUC in the notebook.
Responsible AI	No demographic/protected-attribute variables used	Model is based mainly on operational, geography and product/order information.